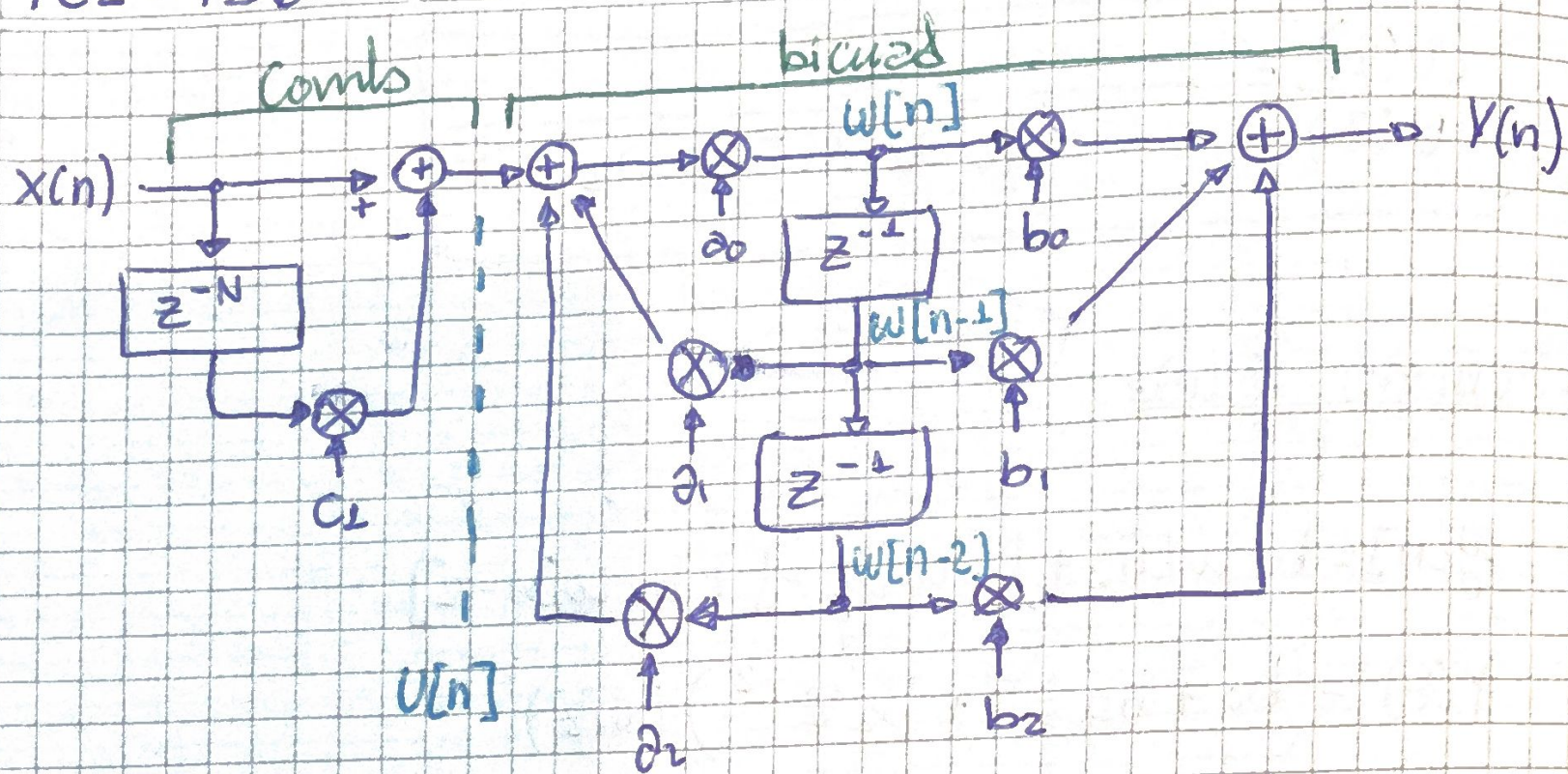




①

Comb

$$U[n] = X[n] - C_1 X[n-N]$$

donde:

$$X[n-N] \xrightarrow{z} z^{-N} X(z)$$

$$U(z) = X(z) - C_1 \cdot z^{-N} X(z) = X(z) [1 - C_1 \cdot z^{-N}]$$

$$\boxed{\frac{U(z)}{X(z)} = 1 - C_1 z^{-N}}$$

biquad

o sumador entrada

$$w[n] = a_0 (U[n] + a_1 w[n-1] + a_2 w[n-2])$$

$$W(z) = a_0 (U(z) + a_1 z^{-1} W(z) + a_2 z^{-2} W(z))$$

$$W(z)(1 - a_0 a_1 z^{-1} - a_0 a_2 z^{-2}) = a_0 U z$$

$$\boxed{\frac{W(z)}{U(z)} = \frac{a_0}{1 - a_0 a_1 z^{-1} - a_0 a_2 z^{-2}}}$$

• Summed Slide

$$y[n] = b_0 w[n] + b_1 w[n-1] + b_2 w[n-2]$$

$$Y(z) = (b_0 + b_1 z^{-1} + b_2 z^{-2}) W(z)$$

$$\boxed{\frac{Y(z)}{W(z)} = b_0 + b_1 z^{-1} + b_2 z^{-2}}$$

Total biquad

$$\boxed{\frac{W(z)}{U(z)} \cdot \frac{Y(z)}{W(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}}$$

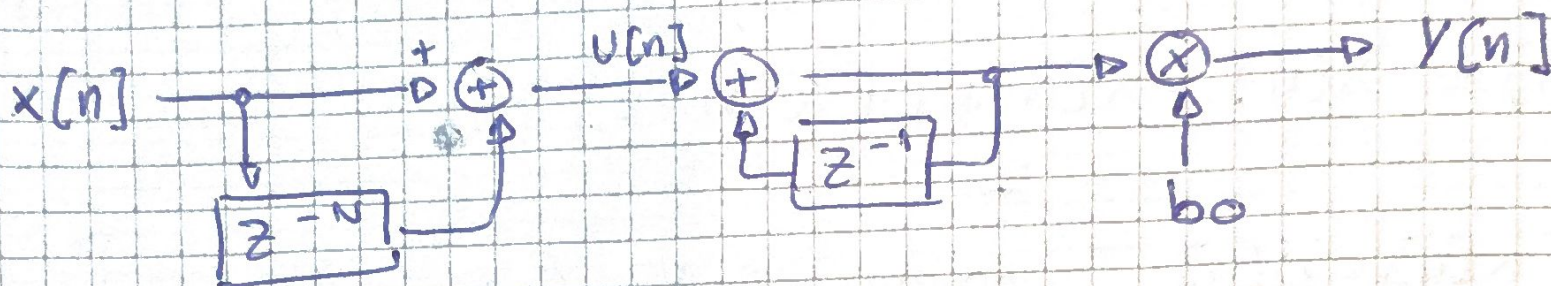
Total (comb + biquad)

$$\boxed{H(z) = \frac{Y(z)}{X(z)} = \frac{U(z)}{X(z)} \cdot \frac{Y(z)}{U(z)} = (1 - c_1 z^{-N}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}}$$

② FILTRO: Media móvil

$$a_0 = 1; a_1 = 1; b_0 = 1/N; c_1 = 1; N = 3, 4, 5$$

$$a_2 = 0; b_1 = 0; b_2 = 0$$



$$M(z) = \frac{1}{N} \cdot \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$Y(z)(1 - z^{-1}) = X(z) \frac{1}{N} (1 - z^{-N})$$

$$Y(z) - z^{-1}Y(z) = \frac{X(z)}{N} - \frac{X(z) \cdot z^{-N}}{N}$$

$$Y[n] - Y[n-1] = \frac{1}{N} (X[n] - X[n-N])$$

$$Y[n] = Y[n-1] + \frac{X[n] - X[n-N]}{N}$$

$$1 - z^{-N} = (1 - z^{-1})(1 + z^{-1} + z^{-2} + \dots + z^{-(N-1)})$$

$$H(z) = \frac{1}{N} \cdot \frac{(1 - z^{-1})(1 + z^{-1} + z^{-2} + \dots + z^{-(N-1)})}{(1 - z^{-1})}$$

$$H(z) = \frac{1}{N} (1 + z^{-1} + \dots + z^{-(N-1)})$$

$$N=3$$

$$H(z) = \frac{1 + z^{-1} + z^{-2}}{3}$$

$$Y[n] = \frac{X[n] + X[n-1] + X[n-2]}{3}$$

Si $x[n] = \delta[n]$ donde $\delta[n] = (1; 0; 0; \dots)$

para $n=0$

$$Y[0] = \frac{X[0] + X[-1] + X[-2]}{3} = \frac{1 + 0 + 0}{3} = \frac{1}{3}$$

para $n=1$

$$Y[1] = \frac{X[1] + X[0] + X[-1]}{3} = \frac{0 + 1 + 0}{3} = \frac{1}{3}$$

para $n=2$

$$Y[2] = \frac{X[2] + X[1] + X[0]}{3} = \frac{0 + 0 + 1}{3} = \frac{1}{3}$$

para $n=3$

$$Y[3] = \frac{X[3] + X[2] + X[1]}{3} = \frac{0 + 0 + 0}{3} = 0$$

$$X[n] = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0, 0 \right)$$

$$h[n] = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

Perz $N=4$

$$H_4(z) = \frac{1 + z^{-1} + z^{-2} + z^{-3}}{4}$$

$$h_4[n] = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$$

Perz $N=5$

$$H_5(z) = \frac{1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}}{5}$$

$$h_5[n] = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right)$$

► FIR o IIR?

FIR : la respuesta al impulso termina despues de una cant. finita de muestras

IIR : la respuesta al impulso continua indefinidamente

En este filtro:

$$h[n] = \begin{cases} \frac{1}{N}, & 0 \leq n \leq N-1 \\ 0 & \text{en los demas casos} \end{cases}$$

El filtro es un FIR.

• La ventaja es que aprovecho la cuenta anterior pero no valgo a sumar casi las mismas números nuevamente

$$Y[n] = Y[n-1] + \frac{x[n] - x[n-N]}{N}$$

• Con un FIR Directo sumaría todo

$$Y[n] = \frac{x[n] + x[n-1] + \dots + x[n-N+1]}{N}$$

$$\Rightarrow h[n] = (1, 1, 1, 1, 1, 1, 1) ?$$

$$H(z) = \frac{1 - z^{-7}}{1 - z^{-1}} \quad (\text{No Dividimos } \times 7)$$

$$N=7, \quad a_0 = a_1 = a_2 = 1, \quad b_0 = 1, \quad a_2 = b_1 = b_2 = 0$$

• Se puede implementar con esa topología pero con $b_0 = 1$

► Polos y Ceros

$$H(z) = \frac{1}{N} \cdot \frac{z^{N-1} + z^{N-2} + \dots + z + 1}{z^{N-1}}$$

$$N=3$$

$$H(z) = \frac{z^2 + z + 1}{3z^2}$$

$$\text{Polo} \rightarrow p_{1,2} = 0$$

$$\text{Cero} \rightarrow z_{1,2} = \frac{-1 \pm \sqrt{3}j}{2}$$

$$N=4$$

$$p_{1,2,3} = 0 \quad z_k = 1 \angle \left(\frac{360^\circ k}{N} \right) \quad k=1,2,3$$

$$z_1 = 1 \angle 90^\circ = -j$$

$$z_2 = 1 \angle 180^\circ = -1$$

$$z_3 = 1 \angle 270^\circ = +j$$

$$N=5$$

$$p_{1,2,3,4} = 0$$

$$z_1 = 1 \angle 72^\circ$$

$$z_2 = 1 \angle 144^\circ$$

$$z_3 = 1 \angle 216^\circ$$

$$z_4 = 1 \angle 288^\circ$$

Mod. y Fase

$$H(z) = \frac{1}{N} \cdot \frac{1 - z^{-N}}{1 - z^{-1}} \rightarrow \frac{1}{N} \cdot \frac{1 - e^{-Nj\omega}}{1 - e^{-j\omega}}$$

$$1 - e^{-j\omega} = e^{-j\omega/2} (e^{+j\omega/2} - e^{-j\omega/2}) = 2je^{-j\omega/2} \sin(\omega/2)$$

$$H(e^{j\omega}) = \frac{1}{N} \cdot \frac{e^{-j\omega N/2} \sin(N\omega/2)}{2je^{-j\omega/2} \sin(\omega/2)}$$

$$H(e^{j\omega}) = e^{-j\omega \frac{(N-1)}{2}} \cdot \frac{\sin(N\omega/2)}{N \sin(\omega/2)}$$

Mod

$$|H(e^{j\omega})| = \left| \frac{\sin(N\omega/2)}{N \sin(\omega/2)} \right|$$

Phase

$$\phi(\omega) = -\omega \frac{N-1}{2} + \begin{cases} 0 & |H(\omega)| > 0 \\ \pi & |H(\omega)| < 0 \end{cases}$$

③ Filtro: Diferenciador

$$a_i \text{ y } b_j ? \rightarrow Y[n] = \frac{X[n] - X[n-2]}{2}$$

$$Y(z) = \frac{1}{2} X(z) - \frac{1}{2} z^{-2} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{2} - \frac{1}{2} z^{-2}$$

$$H(z) = (1 - C_1 z^{-N}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

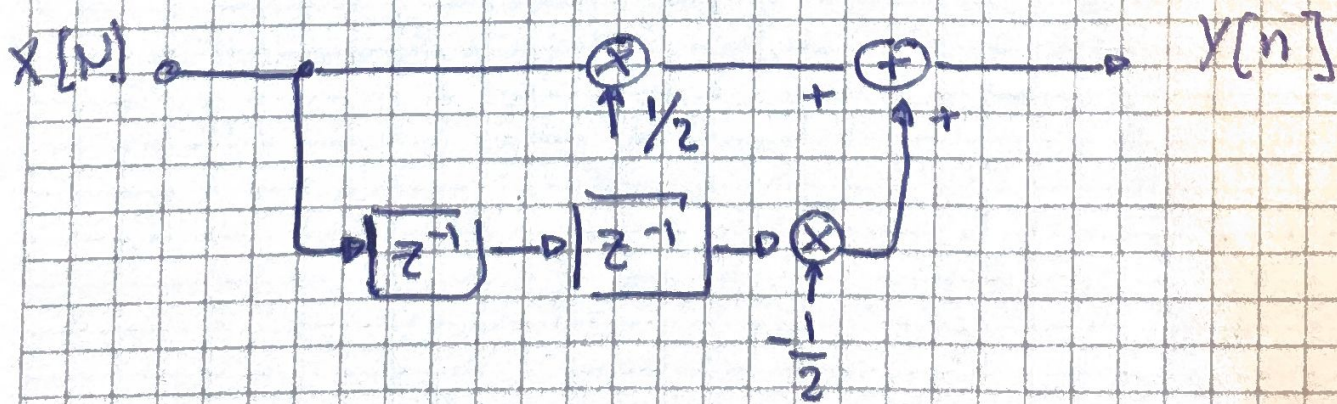
$$C_1 = 0$$

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

$$a_0 = 1 ; a_1 = a_2 = 0 ; b_1 = 0$$

$$H(z) = b_0 + b_2 z^{-2} \quad \left\{ \begin{array}{l} b_0 = \frac{1}{2} \\ b_2 = -\frac{1}{2} \end{array} \right.$$

Diagrama



Polos y Ceros

$$H(z) = \frac{1}{2} (1 - z^{-2}) = \frac{z^2 - 1}{2z^2} = \frac{(z-1)(z+1)}{2z^2}$$

$$z_1 = 1 \quad z_2 = -1 \quad p_{1,2} = 0$$

Mod. Fase

$$H(e^{jn}) = \frac{1}{2} (1 - e^{-jn2}) = \frac{1}{2} e^{-jn} (e^{jn} - e^{-jn})$$

$$\bullet e^{jn} - e^{-jn} = 2j \sin n$$

$$H(e^{jn}) = j e^{-jn} \sin(n)$$

Mod

$$|H(e^{jn})| = |\sin(n)|$$

Fase

$$j = \frac{\pi}{2}$$

$$\varphi(n) = \frac{\pi}{2} - n$$

④ Filtro: eliminaz continuaz.

$$a_0 = 1 ; a_1 = \beta ; b_0 = 1 ; b_1 = -1 ; \beta = 0,9$$

$$H(z) = (1 - a_1 z^{-1}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

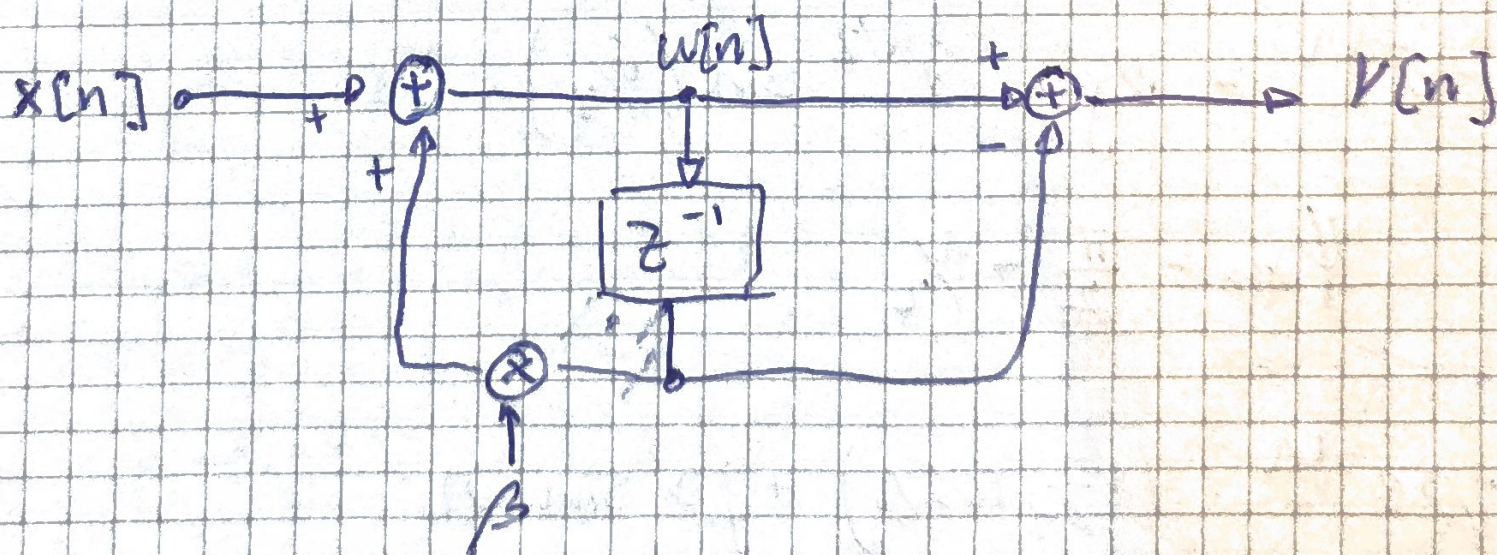
$$H(z) = \frac{1 - z^{-1}}{1 - \beta z^{-1}} = \frac{1 - z^{-1}}{1 - 0,9 z^{-1}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - z^{-1}}{1 - 0,9 z^{-1}}$$

$$Y(z) - \beta z^{-1} Y(z) = X(z) - z^{-1} X(z)$$

$$Y[n] = X[n] - X[n-1] + \beta Y[n-1]$$

Diagrama en Bloques



Polar y Ceros

$$H(z) = \frac{z-1}{z-\beta}$$

$$z = 1$$

$$p = \beta$$

Mod Fase

$$H(e^{jn}) = \frac{1 - e^{-jn}}{1 - \beta e^{-jn}}$$

$$|1 - e^{-jn}|^2 = (1 - e^{-jn})(1 - e^{jn}) = 2 - 2\cos n$$

$$|1 - \beta e^{-jn}|^2 = (1 - \beta e^{jn})(1 - \beta e^{-jn}) = 1 + \beta^2 - 2\beta \cos n$$

Mod

$$|H(e^{jn})| = \sqrt{\frac{2 - 2\cos n}{1 + \beta^2 - 2\beta \cos n}}$$

Fase

$$1 - e^{-jn} = 2j e^{-jn/2} \sin(n/2)$$

$$\varphi_{num} = \frac{\pi}{2} - \frac{n}{2}$$

$$1 - \beta e^{-jn} = (1 - \beta \cos n) + j\beta \sin n$$

$$\varphi_{den} = \operatorname{tg}^{-1} \left(\frac{\beta \sin(n)}{1 - \beta \cos(n)} \right)$$

$$\varphi_r = \frac{\pi}{2} - \frac{\Omega}{2} - \tan^{-1} \left(\frac{\beta \sin(\Omega)}{1 - \beta \cos(\Omega)} \right)$$

$$\Rightarrow \beta? \text{ para } \Omega = 0,1 \cdot \pi \text{ sea } < 3 \text{ dB} \text{ e } \Omega = \pi$$

$$20 \log \left(\frac{|H(e^{j\Omega_c})|}{|H(e^{j\pi})|} \right) = -3$$

$$\frac{|H(e^{j\Omega_c})|^2}{|H(e^{j\pi})|^2} = 10^{-3/10}$$

$$|H(e^{j\Omega_c})|^2 = \frac{2 - 2 \cos \Omega_c}{1 + \beta^2 - 2\beta \cos \Omega_c}$$

$$\cos(\pi) = -1$$

$$|H(e^{j\pi})|^2 = \frac{2 - 2(-1)}{1 + \beta^2 - 2\beta(-1)} = \frac{2+2}{1+\beta^2+2\beta} = \frac{4}{\beta^2+2\beta+1}$$

$$c = \cos \Omega_c \quad \text{y} \quad q = 10^{-3/10}$$

$$\frac{2-2c}{\beta^2-2\beta c+1} \cdot \frac{\beta^2+2\beta+1}{4} = q$$

$$(1-c)(\beta+1)^2 = 2q(\beta^2-2\beta c+1)$$

$$\beta^2+2\beta+1 - q\beta^2 - 2c\beta - c = 2q\beta^2 - 4qc\beta + 2q$$

$$(2q-(1-c))\beta^2 - (4qc+2(1-c))\beta + (2q-(1-c)) = 0$$

A

B

C

$$Q \approx 0,501187$$

$$C \approx 0,951057$$

$$A = 0,9534$$

$$B = -2,0045$$

$$C = 0,9534$$

$$\beta_1 \approx 1,3753$$

$$\beta_2 \approx 0,7271$$

Como $0 < |\beta| < 1$

$$\beta \approx 0,7271025814$$

5) Filtro Peine

$$C_1 = -1 \quad N = 8$$

$$a_0 = b_0 = 1 \quad a_1 = a_2 = b_2 = b_1 = 0$$

$$H(z) = (1 - C_1 z^{-N}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

$$H(z) = 1 + 1 \cdot z^{-8}$$

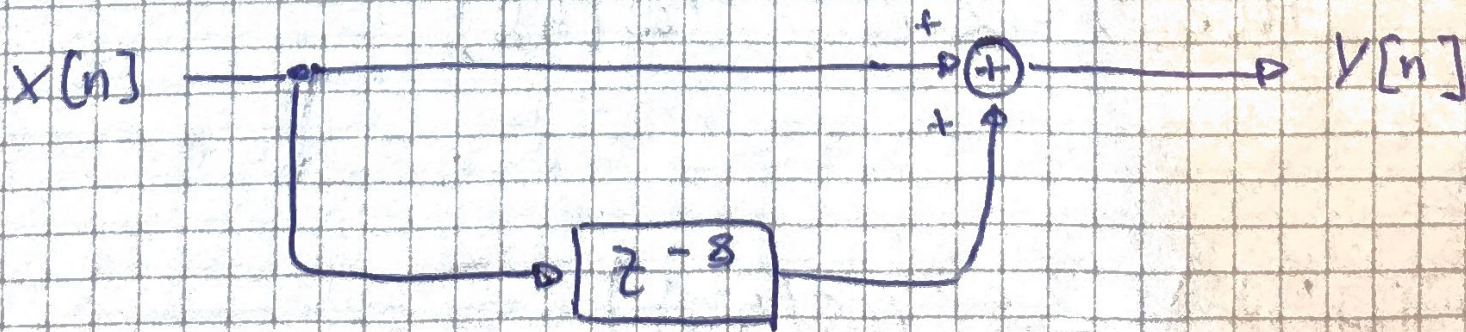
$$H(z) = 1 + z^{-8}$$

$$H(z) = \frac{Y(z)}{X(z)} = 1 + z^{-8}$$

$$Y(z) = X(z) + z^{-8} X(z)$$

$$Y[n] = X[n] + X[n-8]$$

Diagrama en Bloques



Polar y Conjug

$$H(z) = \frac{z^8 + 1}{z^8}$$

$$P_{1,2,\dots,8} = 0$$

$$z^8 + 1 = 0 \Rightarrow z^8 = -1$$

$$z_k = e^{j(2k+1)\pi/8} \quad k=0, 1, \dots, 7$$

$$\theta_k = \frac{(2k+1)180^\circ}{8}$$

$$z_0 = 1 \angle 22,5^\circ$$

$$z_5 = 1 \angle 247,5^\circ$$

$$z_1 = 1 \angle 67,5^\circ$$

$$z_6 = 1 \angle 292,5^\circ$$

$$z_2 = 1 \angle 112,5^\circ$$

$$z_7 = 1 \angle 337,5^\circ$$

$$z_3 = 1 \angle 157,5^\circ$$

$$z_4 = 1 \angle 202,5^\circ$$

Mod f2se

$$H(e^{j\Omega}) = 1 + e^{-j8\Omega} = e^{-j4\Omega} (e^{j4\Omega} + e^{-j4\Omega})$$

$$e^{j\theta} + e^{-j\theta} = 2 \cos \theta$$

$$H(e^{j\Omega}) = 2 e^{-j4\Omega} \cos(4\Omega)$$

Mod

$$|H(e^{j\Omega})| = 2 |\cos(4\Omega)|$$

Fase

$$\varphi = -4\Omega + \begin{cases} 0 & \cos(4\Omega) > 0 \\ \pi & \cos(4\Omega) < 0 \end{cases}$$

$$f_{\text{sampling}} = 8 \text{ kHz}$$

$$\Omega = 2\pi \cdot \frac{f}{f_s} \rightarrow 0 \leq f \leq \frac{f_s}{2} = 4000 \text{ Hz}$$

Picos $\rightarrow \underbrace{|\cos(4\Omega)| = 1}$

$$4\Omega = k\pi \rightarrow \Omega_{\text{pico}} = \frac{k\pi}{4}$$

$$\frac{2\pi \cdot f_{\text{pico}}}{f_s} = \frac{k\pi}{4}$$

$$f_{\text{pico}} = \frac{f_s k}{8}$$

$$f_{\text{pico}} = 0, 1\text{k}, 2\text{k}, 3\text{k}, 4\text{k} \text{ [Hz]}$$

$$|H| = 2 \rightarrow 20 \log(2) = 6,02 \text{ dB}$$

Nulls $\rightarrow \underbrace{\cos(4\Omega) = 0}$

$$4\Omega = \frac{\pi}{2} + k\pi \rightarrow \Omega_{\text{nulo}} = \frac{(2k+1)\pi}{8}$$

$$f_{\text{nulo}} = \frac{(2k+1)f_s}{16} \quad f_{\text{nulo}} = 500, 1,5\text{k}, 2,5\text{k}, 3,5\text{k} \text{ [Hz]}$$

Bonus

$$H(s) = \frac{s - \omega_0}{s + \omega_0}$$

$$|H(j\omega)| = \frac{\sqrt{\omega^2 + \omega_0^2}}{\sqrt{\omega^2 + \omega_0^2}} = 1$$

$$s = \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}} = k \frac{1 - z^{-1}}{1 + z^{-1}} \quad k = 2f_s$$

$$H_d(z) = \frac{k \frac{1 - z^{-1}}{1 + z^{-1}} - \omega_0}{k \frac{1 - z^{-1}}{1 + z^{-1}} + \omega_0} = \frac{k - k z^{-1} - \omega_0 - \omega_0 z^{-1}}{k - k z^{-1} + \omega_0 + \omega_0 z^{-1}}$$

$$H_d(z) = \frac{(k - \omega_0) - (k + \omega_0) z^{-1}}{(k + \omega_0) - (k - \omega_0) z^{-1}} = \frac{\left(\frac{k - \omega_0}{k + \omega_0}\right) - z^{-1}}{1 - \left(\frac{k - \omega_0}{k + \omega_0}\right) z^{-1}}$$

$$\alpha = \frac{k - \omega_0}{k + \omega_0}$$

$$\Rightarrow H_d(z) = \frac{\alpha - z^{-1}}{1 - \alpha z^{-1}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\alpha - z^{-1}}{1 - \alpha z^{-1}} \Rightarrow Y(z) = \alpha X(z) - z^{-1} X(z)$$

$$Y(z) - \alpha z^{-1} Y(z) = \alpha X(z) - z^{-1} X(z)$$

$$Y(z) = \alpha z^{-1} Y(z) + \alpha X(z) - z^{-1} X(z)$$

$$y[n] = \alpha y[n-1] + \alpha x[n] - x[n-1]$$

